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Question 1 [C01] [10 Points]

You are a software developer at the **Dhaka Power Distribution Company (DPDC)**, assigned to create a **Java program** that calculates customers' **monthly electricity bills** automatically. The system should take two inputs – the number of **units consumed** and the number of **days the payment is late**. The billing rate is **5 BDT per unit** for the first 100 units, **7 BDT per unit** for the next 200 units, and **10 BDT per unit** for any usage above 300 units. If the payment is delayed by **up to 14 days**, a **10% penalty** is added to the total bill, and if it's **more than 14 days late**, an additional **2% penalty per day** applies for every extra day of delay.

Your program should calculate and display the **total amount payable** clearly.

Input	Output
The number of units consumed = 320 units Days of late payment = 20 days	Total payable amount = 2562 BDT
<u>Explanation:</u> First 100 units = $100 \times 5 = 500$ BDT Next 200 units = $200 \times 7 = 1400$ BDT Remaining 20 units = $20 \times 10 = 200$ BDT Base Bill = $500 + 1400 + 200 = 2100$ BDT Days late = 20 Extra days = $20 - 14 = 6$ days Total penalty rate = $10\% + (6 \times 2\%) = 10\% + 12\% = 22\%$ Penalty = $22\% \text{ of } 2100 = 0.22 \times 2100 = 462$ BDT Total Bill = Base Bill + Penalty Total Bill = $2100 + 462 = 2562$ BDT	